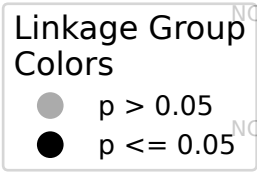


diamond: 10137 orthologs on 4167 Rphi scaffolds and 12 Cgig scaffolds.



Rphi vs Cphi groupings

Rphi vs Cgig FET, whole chroms.

RphRph scaffolds

	Rphi scaf	Cgig scaf	p (FET)	Count
0	NW_026860903.1	NC_088859.1	4.30e-13	21
1	NW_026865571.1	NC_088858.1	1.09e-11	20
2	NW_026865484.1	NC_088858.1	6.64e-10	19
3	NW_026861563.1	NC_088858.1	7.31e-11	19
4	NW_026862663.1	NC_088859.1	1.66e-08	18
5	NW_026861019.1	NC_088858.1	2.57e-08	18
6	NW_026853802.1	NC_088859.1	5.08e-09	16
7	NW_026851633.1	NC_088860.1	1.49e-11	16
8	NW_026863896.1	NC_088855.1	2.63e-05	15
9	NW_026858503.1	NC_088856.1	1.61e-13	15
10	NW_026855197.1	NC_088855.1	1.06e-04	15
11	NW_026852258.1	NC_088858.1	1.06e-06	15
12	NW_026852008.1	NC_088859.1	4.54e-07	15
13	NW_026865967.1	NC_088859.1	2.77e-06	14
14	NW_026863541.1	NC_088858.1	7.32e-08	14
15	NW_026859933.1	NC_088859.1	9.22e-05	14
16	NW_026854960.1	NC_088859.1	2.16e-07	14
17	NW_026852705.1	NC_088859.1	2.77e-06	14
18	NW_026861864.1	NC_088855.1	2.48e-05	13
19	NW_026861084.1	NC_088855.1	1.59e-04	13
20	NW_026859561.1	NC_088857.1	3.84e-08	13
21	NW_026853170.1	NC_088859.1	1.40e-06	13
22	NW_026852000.1	NC_088854.1	6.37e-07	13
23	NW_026862783.1	NC_088859.1	1.02e-04	12
24	NW_026859790.1	NC_088858.1	3.62e-06	12
25	NW_026855294.1	NC_088855.1	3.68e-02	12
26	NW_026867313.1	NC_088856.1	7.66e-09	11
27	NW_026867092.1	NC_088859.1	5.93e-05	11
28	NW_026863645.1	NC_088860.1	1.82e-06	11
29	NW_026861840.1	NC_088857.1	7.33e-04	11
30	NW_026857770.1	NC_088855.1	1.88e-02	11
31	NW_026857221.1	NC_088855.1	8.48e-04	11
32	NW_026853905.1	NC_088858.1	2.54e-05	11
33	NW_026866837.1	NC_088858.1	1.71e-03	10
34	NW_026866451.1	NC_088859.1	3.64e-03	10
35	NW_026865527.1	NC_088855.1	5.21e-04	10
36	NW_026864352.1	NC_088859.1	1.88e-02	10
37	NW_026863691.1	NC_088856.1	1.16e-06	10
38	NW_026863406.1	NC_088856.1	1.16e-06	10
39	NW_026862699.1	NC_088858.1	1.71e-03	10
40	NW_02685998.1	NC_088859.1	3.85e-04	10
41	NW_026859757.1	NC_088859.1	3.85e-04	10
42	NW_026859620.1	NC_088858.1	1.71e-03	10
43	NW_026859418.1	NC_088858.1	1.78e-04	10
44	NW_026852806.1	NC_088855.1	4.90e-03	10
45	NW_026852468.1	NC_088858.1	1.71e-03	10
46	NW_026867154.1	NC_088858.1	1.25e-03	9
47	NW_026866838.1	NC_088860.1	2.04e-05	9
48	NW_026866080.1	NC_088858.1	1.09e-02	9
49	NW_026865813.1	NC_088858.1	5.23e-02	9
50	NW_026864957.1	NC_088858.1	1.25e-03	9
51	NW_026864022.1	NC_088857.1	2.08e-04	9
52	NW_026863843.1	NC_088854.1	1.26e-02	9
53	NW_026861844.1	NC_088859.1	2.50e-03	9
54	NW_026861602.1	NC_088858.1	5.23e-02	9
55	NW_026859085.1	NC_088859.1	2.50e-03	9
56	NW_026857092.1	NC_088854.1	1.45e-03	9
57	NW_026857461.1	NC_088855.1	3.28e-03	9
58	NW_026856555.1	NC_088859.1	2.50e-03	9
59	NW_026856137.1	NC_088858.1	1.09e-02	9
60	NW_026855828.1	NC_088859.1	2.15e-02	9
61	NW_026855648.1	NC_088859.1	2.50e-03	9
62	NW_026855204.1	NC_088855.1	2.81e-02	9
63	NW_026854713.1	NC_088855.1	3.28e-03	9

	Rphi scaf	Cgig scaf	p (FET)	Count
0	NW_026860903.1 : 0-end	NC_088859.1 : 0-end	4.30e-13	21
1	NW_026865571.1 : 0-end	NC_088858.1 : 0-end	1.09e-11	20
2	NW_026865484.1 : 0-end	NC_088858.1 : 0-end	6.64e-10	19
3	NW_026861563.1 : 0-end	NC_088858.1 : 0-end	7.31e-11	19
4	NW_026862663.1 : 0-end	NC_088859.1 : 0-end	1.66e-08	18
5	NW_026861019.1 : 0-end	NC_088858.1 : 0-end	2.57e-08	18
6	NW_026853802.1 : 0-end	NC_088859.1 : 0-end	5.08e-09	16
7	NW_026851633.1 : 0-end	NC_088860.1 : 0-end	1.49e-11	16
8	NW_026863896.1 : 0-end	NC_088855.1 : 0-end	2.63e-05	15
9	NW_026858503.1 : 0-end	NC_088856.1 : 0-end	1.61e-13	15
10	NW_026855197.1 : 0-end	NC_088855.1 : 0-end	1.06e-04	15
11	NW_02685258.1 : 0-end	NC_088858.1 : 0-end	1.06e-06	15
12	NW_026852008.1 : 0-end	NC_088859.1 : 0-end	4.54e-07	15
13	NW_026865967.1 : 0-end	NC_088859.1 : 0-end	2.77e-06	14
14	NW_026863541.1 : 0-end	NC_088858.1 : 0-end	7.32e-08	14
15	NW_026859933.1 : 0-end	NC_088859.1 : 0-end	9.22e-05	14
16	NW_026854960.1 : 0-end	NC_088859.1 : 0-end	2.16e-07	14
17	NW_026852705.1 : 0-end	NC_088859.1 : 0-end	2.77e-06	14
18	NW_026861864.1 : 0-end	NC_088855.1 : 0-end	2.48e-05	13
19	NW_026861084.1 : 0-end	NC_088855.1 : 0-end	1.59e-04	13
20	NW_026859561.1 : 0-end	NC_088857.1 : 0-end	3.84e-08	13
21	NW_026853170.1 : 0-end	NC_088859.1 : 0-end	1.40e-06	13
22	NW_026852000.1 : 0-end	NC_088854.1 : 0-end	6.37e-07	13
23	NW_026862783.1 : 0-end	NC_088859.1 : 0-end	1.02e-04	12
24	NW_026859790.1 : 0-end	NC_088858.1 : 0-end	3.62e-06	12
25	NW_026855294.1 : 0-end	NC_088855.1 : 0-end	3.68e-02	12
26	NW_026867313.1 : 0-end	NC_088856.1 : 0-end	7.66e-09	11
27	NW_026867092.1 : 0-end	NC_088859.1 : 0-end	5.93e-05	11
28	NW_026863645.1 : 0-end	NC_088860.1 : 0-end	1.82e-06	11
29	NW_026861840.1 : 0-end	NC_088857.1 : 0-end	7.33e-04	11
30	NW_026857770.1 : 0-end	NC_088855.1 : 0-end	1.88e-02	11
31	NW_026857221.1 : 0-end	NC_088855.1 : 0-end	8.48e-04	11
32	NW_026853905.1 : 0-end	NC_088858.1 : 0-end	2.54e-05	11
33	NW_026866837.1 : 0-end	NC_088858.1 : 0-end	1.71e-03	10
34	NW_026866451.1 : 0-end	NC_088859.1 : 0-end	3.64e-03	10
35	NW_026865227.1 : 0-end	NC_088855.1 : 0-end	5.21e-04	10
36	NW_026864352.1 : 0-end	NC_088859.1 : 0-end	1.88e-02	10
37	NW_026863691.1 : 0-end	NC_088856.1 : 0-end	1.16e-06	10
38	NW_026863406.1 : 0-end	NC_088856.1 : 0-end	1.16e-06	10
39	NW_026862689.1 : 0-end	NC_088858.1 : 0-end	1.71e-03	10
40	NW_026859998.1 : 0-end	NC_088859.1 : 0-end	3.85e-04	10
41	NW_026859757.1 : 0-end	NC_088859.1 : 0-end	3.85e-04	10
42	NW_026859620.1 : 0-end	NC_088858.1 : 0-end	1.71e-03	10
43	NW_026859418.1 : 0-end	NC_088858.1 : 0-end	1.78e-04	10
44	NW_026852806.1 : 0-end	NC_088855.1 : 0-end	4.90e-03	10
45	NW_026852468.1 : 0-end	NC_088858.1 : 0-end	1.71e-03	10
46	NW_026867154.1 : 0-end	NC_088858.1 : 0-end	1.25e-03	9
47	NW_026866838.1 : 0-end	NC_088860.1 : 0-end	2.04e-05	9
48	NW_026866080.1 : 0-end	NC_088858.1 : 0-end	1.09e-02	9
49	NW_026865813.1 : 0-end	NC_088858.1 : 0-end	5.23e-02	9
50	NW_026864957.1 : 0-end	NC_088858.1 : 0-end	1.25e-03	9
51	NW_026864022.1 : 0-end	NC_088857.1 : 0-end	2.08e-04	9
52	NW_026863843.1 : 0-end	NC_088854.1 : 0-end	1.26e-02	9
53	NW_026861844.1 : 0-end	NC_088859.1 : 0-end	2.50e-03	9
54	NW_026861602.1 : 0-end	NC_088858.1 : 0-end	5.23e-02	9
55	NW_026859085.1 : 0-end	NC_088859.1 : 0-end	2.50e-03	9
56	NW_026857902.1 : 0-end	NC_088854.1 : 0-end	1.45e-03	9
57	NW_026857461.1 : 0-end	NC_088855.1 : 0-end	3.28e-03	9
58	NW_026856555.1 : 0-end	NC_088859.1 : 0-end	2.50e-03	9
59	NW_026856137.1 : 0-end	NC_088858.1 : 0-end	1.09e-02	9
60	NW_026855828.1 : 0-end	NC_088859.1 : 0-end	2.15e-02	9
61	NW_026855648.1 : 0-end	NC_088859.1 : 0-end	2.50e-03	9
62	NW_026855204.1 : 0-end	NC_088855.1 : 0-end	2.81e-02	9
63	NW_026854713.1 : 0-end	NC_088855.1 : 0-end	3.28e-03	9

(a) depicts the chromosomes/scaffolds of Rphi (x-axis) plotted against the chromosomes/scaffolds of Cgig (y-axis). Each dot in the plot represents an ortholog, specifically a reciprocal best diamond blastp match between two species. The unit of the x- and y-axes are the number of orthologous proteins between these two species: 10137 orthologs found between 4167 Rphi scaffolds and 12 Cgig scaffolds. If there are chromosome breaks, Fisher's exact test is used to calculate the significance of the interactions between the sub-chromosomal pieces. Otherwise Fisher's exact test is calculated on whole chromosomes. The opacity of the dots depict the significance from Fisher's exact test. Dots that are a solid color are in cells with a FET p-value less than or equal to 0.05. Dots that are translucent are in cells with a FET p-value greater than 0.05. The Dx and Dy values depict places where there may be sudden breaks in synteny. See the supplementary information the following paper for more information on Dx and Dy: Simakov, Oleg, et al. Deeply conserved synteny resolves early events in vertebrate evolution. Nature Ecology & Evolution 4.6 (2020): 820-830. (b) depicts the same information as panel a, but it is plotted in the organisms' chromosome basepair coordinates rather than gene index. This is useful for visualizing gene-poor regions of the chromosomes. (c) shows which gene groups are most prevalent in each chromosome pair. The FET p-value in this table corresponds to the whole-chromosome FET p-value, and is not a FET value of the correlation between the chromosome pair and the gene group. (d) shows chromosome-scale significance values. This table shows the same information as c, but does not show the gene group. (e) shows the FET p-values for each chromosome pair. (f) shows the FET p-values for each chromosome pair. (g) shows the FET p-values for each chromosome pair.

	NW_026865278.1	NC_088554.1	0.99 - 0.8	73	NW_026865278.1	NC_088554.1	0.99 - 0.8
74	NW_026865278.1	NC_088554.1	0.99 - 0.8	74	NW_026865278.1	NC_088554.1	0.99 - 0.8
75	NW_026865278.1	NC_088554.1	0.99 - 0.8	75	NW_026865278.1	NC_088554.1	0.99 - 0.8